Polynomial PEPS Approximation of Gapped Square-Grid Ground States

Exact Encoded Frames, Whole-Group Truncation, and Conditional Resource Bounds

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Abstract

We examine the September 24, 2026 source claiming polynomial-bond PEPS approximation, in global vector norm, for unique uniformly gapped ground states on open square grids. All fourteen text files in its build tree were read, including the information, adaptive-patch, distributed-compression, frame, geometry, and assembly proofs. This is a scientific read and a dependency audit, not a certification of that global claim. In particular, the companion sublinear-collar theorem and its full area-law argument remain unverified dependencies; two relevant companion proof files were read in full. We give complete finite-dimensional proofs of an exact encoded-range and leakage account, a sharper whole-group truncation estimate $\frac{1}{2}(\sum_j \tau_j^2/k_j)^{1/2}$, a reduced coefficient budget, nonuniform Gaussian slot allocation, and a sharp normalization bound for column extraction from a possibly nonpositive operator network. Physical Gram matrices and the omitted residual component are retained explicitly. The refinements improve particular bounds under their stated hypotheses; they establish neither a new general two-dimensional area law nor an efficient ground-state algorithm. Contribution assessment: C2/5, complete restricted refinements and source-specific error accounting with substantial unverified global dependencies and no claim of new global PEPS theory.

Contents

1 Question, Domain, and Status of the Claim **2**
1.1 What this report does and does not establish 3

2 Read Scope and Primary Context **3**
2.1 Current primary literature and non-equivalent conclusions 3

3 The Source's Dependency Chain **4**
3.1 Companion positive replacement: what can actually be checked 4
3.2 Amplification and its remaining inputs 5

4 Information Reset and Adaptive Patches **5**
4.1 Why a one-sided filter estimate would be insufficient 5
4.2 Conditional information is not monotone under conditioning 5
4.3 The noncommuting variational patch construction 6

5 Exact Encoded Frames and Their Actual Domain **7**

6 Sharper Whole-Group Tensor Truncation **8**
6.1 Rank allocation and operator branches 9

7 Rewrites, Pair Effects, and Density Compression **9**
7.1 The exact rewrite before approximation 9
7.2 Common garbage for effects 10
7.3 The dimension-free Gaussian calculation 10
7.4 A precise nonuniform resource extension 11

8 Ownership Geometry and Global Assembly **12**
8.1 Bulk bands and point repairs 12
8.2 Errors and routing 13

9 Column Extraction Without Positivity **13**

10 Physical Metrics and Honest Candidate Checks **14**

11 Gap Audit, Benefits, Costs, and Further Work **15**
11.1 Invalid shortcuts excluded by the analysis 15
11.2 Benefits of the restricted reformulation 15
11.3 Costs and limitations 15
11.4 Concrete extensions and their obstacles 15

12 Contribution and Evidence Contract **16**

1 Question, Domain, and Status of the Claim

Let $\Lambda_L = \{1, \dots, L\}^2$ have open nearest-neighbor edges and physical space $\mathcal{H}_L = \bigotimes_{x \in \Lambda_L} \mathbb{C}^q$, with fixed $q \geq 2$. The source [1] considers a Hermitian Hamiltonian

$$H = \sum_x h_x + \sum_{\{x,y\}} h_{xy}, \qquad \|h_x\|, \|h_{xy}\| \leq J, \qquad H - E_0 I \geq \Delta(I - P_\Omega), \quad P_\Omega = |\Omega\rangle\langle\Omega|, \tag{1}$$

where $\|\Omega\| = 1$ and $q, J, \Delta > 0$ do not vary with L. The last inequality includes uniqueness. It does not assume a gap for every subregion Hamiltonian. Frustration freeness, translation invariance, and a gapped interpolation to a product state are not hypotheses.

The claimed conclusion is existence of a nonzero PEPS Ψ_L on the *original* square grid, with bond dimension at most CL^c, such that

$$\min_{|z|=1} \|z\Psi_L / \|\Psi_L\| - \Omega\| \leq L^{-1}. \tag{2}$$

The constants may depend on q, J, Δ, not on the particular volume. Virtual bonds are coordinate pairings of unnormalized maximally entangled vectors; the local PEPS maps need not be isometries. This convention matters when comparing Frobenius norms and physical operator norms.

The construction is existential. Arbitrary private Hilbert spaces, unknown exact copies of Ω, a nonconstructively selected Gaussian realization, and a nonconstructively selected column are allowed in its intermediate representation. Polynomial bond dimension alone proves no polynomial algorithm to find tensors, contract them, certify their entries, or describe them at bounded bit precision.

1.1 What this report does and does not establish

The local propositions below are proved for the explicitly stated finite spaces, projectors, tensor groups, and coefficient budgets. They can be used without accepting the source's global theorem. Whenever the source's adaptive patches or owner geometry are inputs, those inputs are named. An unresolved proof dependency is not a demonstrated counterexample. Conversely, finding no verified error in selected manipulations is not evidence that every analytic or geometric quantifier has been discharged.

The applicability record assigns L4 because exact synthesis/analysis, retained Gram operators, and leakage have direct uses here. L4 measures relevance of the private methods, not the strength of a realized scientific contribution. The realized assessment remains C2.

2 Read Scope and Primary Context

The literal source scope comprises the complete fourteen text files in the pinned M535 build tree: `main.tex`, `references.bib`, two figure-source files, the history and introduction, and sections 01–08. The total is 4543 lines under the line-count convention recorded in the sidecar. In particular the six long proof files were not replaced by their abstracts: information 786 lines, patches 604, compression 611, frames 570, geometry 681, and assembly 214. The interface file has 111 lines. Every line range and pre-read SHA-256 is preserved in the read ledger.

In the distinct companion source [2], the complete files `03-quasilocal.tex` (534 lines) and `09-amplification.tex` (583 lines) were read. The companion's initial marginal-tail proof, conditional/skew estimates, replicas, transport, comparators, scanner, and final area-law geometry were *not* read in full for this report. Thus the companion collar input is not independently established here.

Actual private proofs, rather than titles alone, were read in the volume-2 frame chapter (lines 90–229), the boundary-local frame report TR-rev-1-old-tr248 (200–317 and 390–446), and the entire volume-21 physical-metric chapter. The three routing maps were consulted only to select these proofs. Their exact paths, ranges, and hashes are in the confidential local evidence; no private content was submitted to public search. These methods provide the algebra in Sections 5 and 10, not the missing two-dimensional information theory.

2.1 Current primary literature and non-equivalent conclusions

Primary records were refreshed on October 8, 2026. Huang's current version is dated June 25, 2026, with a 2026 IEEE publication [3]. Its two-dimensional conclusion concerns approximation of local properties under a Rényi area law with index below one; its stated bond dependence is $\exp(O(1/\delta))$. It is not the global vector-norm result (2) under only (1).

Anshu–Arad–Gosset prove an area law for locally gapped, frustration-free two-dimensional systems and a vertical cut [4]. Arad–Firanko–Jain assume a suitable approximate ground-space projector; their available two-dimensional applications retain frustration-free/local-gap restrictions [5]. Neither supplies the unrestricted companion theorem merely by citation.

Molnár–Schuch–Verstraete–Cirac give Gibbs-state PEPS bounds and a quasipolynomial ground-state conclusion under a density-of-states condition [6]. Ge–Eisert demonstrate that general area-law states need not have efficient classical descriptions [7]. Their obstruction does not refute the more restricted class of unique gapped local ground states. It does refute an inference from entropy alone without the extra structure. The earlier PEPS problem survey [8] supplies historical context, not an assertion that its problem list remains unresolved after all subsequent work.

3 The Source's Dependency Chain

The source's route is considerably stronger than "area law implies PEPS." It has the following distinct steps, each with its own norm and quantifiers.

1. An imported sublinear collar estimate and its amplification give coarse mutual-information bounds. An ordered geometric tiling turns them into conditional information bounds for arbitrary unions of cells.

2. A Gaussian-filter reset, using the full gap twice, upgrades small information costs to inverse powers of L across logarithmic collars.

3. An area-law input and a noncommuting variational filter give adaptive patch projectors at radius $O(\log L)$. Their complexity is a sum of *inside* ranks at different radii, not the rank of a fixed outer projector.

4. Tagged encoders replace junction holes by exact partial isometries. Owner changes become contractions involving a bounded number of parties, polynomially many terms, and normalized pair resources/effects.

5. A density-level Gaussian compression eliminates unrestricted private dimensions while retaining a bounded-lifetime interaction graph.

6. A nonzero column of the resulting operator network gives a vector approximation. A dyadic routing with constant congestion gives a PEPS on the original grid.

The conclusions of the last two steps are conditional on the first four. Polynomially many gates are not enough: each party must participate in only a bounded number of nonprivate gates over its *entire lifetime*. An $O(\log L)$ participation count would feed into polynomial coefficient budgets exponentially and can destroy the intended polynomial bound.

3.1 Companion positive replacement: what can actually be checked

Write $H = \sum_i h_i$ with bounded anchor multiplicity. The companion constructs a real even compact spectral cutoff χ, with $\chi(0) = 1$ and support in $(-\Delta/2, \Delta/2)$, whose inverse Fourier kernel has $|f(t)| \le C \exp(-c|t|^\alpha)$, $\alpha = p/(p+1) < 1$. With $\tau_t(A) = e^{itH} A e^{-itH}$, put

$$M_i = \int f(t)\tau_t(h_i)\, dt - \langle \Omega, h_i \Omega \rangle\, I, \quad k_i = |M_i|/c_*, \quad c_* \ge \max_i \|M_i\|\,. \tag{3}$$

In an energy eigenbasis $M_i\Omega = 0$, and summing gives $\sum_i M_i = H - E_0 I$. Thus $0 \le k_i \le I$, $k_i\Omega = 0$, and

$$\sum_i k_i \ge c_*^{-1}(H - E_0 I) \ge (\Delta/c_*)(I - P_\Omega).$$

This is a valid order argument: it does not require the M_i to commute. Lieb–Robinson localization gives a stretched-exponential tail for M_i. For conditional expectation $E_{i,l}$ onto an anchor ball,

$$\| |M_i| - |E_{i,l} M_i| \| \le \left\| M_i^2 - (E_{i,l} M_i)^2 \right\|^{1/2} \le (2c_* \|M_i - E_{i,l} M_i\|)^{1/2}.$$

The exponent constant deteriorates, but the power α persists. The expectation fixes the local absolute value and is contractive, so the same tail, with a factor two, controls $k_i - E_{i,l} k_i$. This does not assert that the original Hamiltonian is frustration free or that the truncated $E_{i,l} k_i$ annihilate Ω.

The companion separately handles truncation by radii $r_i = \max(r_0, \lfloor d(a_i, S_0)/2 \rfloor)$ with $r_0 = C(\log n)^2$. Summing tails costs $|S_0|$, not $|\Lambda|$. If $\left\| \widetilde{H} - H_F \right\| \le \varepsilon_n \le g/4$, positivity and min–max give a

unique ground vector of $\widetilde{H}$, a gap at least $g/2$, and phase distance at most $2\sqrt{\varepsilon_n/g}$ from Ω. The exact ground state changes; it is not silently reused as a zero mode.

3.2 Amplification and its remaining inputs

For $G_i = (I - k_i)^{1/2}$, $K_i = k_i^{1/2}$, the unital channel $\mathcal{E}_i(B) = G_i B G_i + K_i B K_i$ satisfies $\mathcal{E}_i(B)\Omega = G_i B \Omega$. The companion localizes *coupled realizations* of Poisson products using these channels. It bounds the expectation of a norm, not only the norm of an averaged operator. Positivity and the gap give

$$\mathbb{E}\,\|D(I - P_\Omega)\zeta\|^2 \le e^{-gt}\,\|(I - P_\Omega)\zeta\|^2\,.$$

The unitality is indispensable: $B \mapsto G_i B G_i$ alone need not fix a disjoint algebra. The read proof controls spatial oscillations by a weighted kernel with bounded row sum and then removes distant clocks.

For a collar $S_0 = X \cup Y$ with three cuts of size $O(n)$, amplification assumes $I_\Omega(X : \Lambda \setminus S_0) \le Cn^{1-10^{-5}}$. Typical spectral projections for X, Y, S_0 additionally use the companion marginal-tail lemma. A restoring operator has norm $\exp((I + 3n^{3/5})/2)$; its adjoint error has no such factor. Poisson suppression, localization, and a common good realization give both forward and adjoint estimates before polar completion. The resulting collar radius is

$$r \le C_M n^{(1-10^{-5})/\alpha}, \qquad \alpha > 1 - 10^{-6}, \qquad n^{1-10^{-5}} + 2r = o(n^{1-2\cdot 10^{-6}}).$$

This supplies $I(X : J) \le C_M n^{-M}$ for remote J, using continuity with only $\log\dim\mathcal{H}_X = O(n^2)$. The exponents are compatible, but their small separation can require enormous thresholds. Reading this proof does not establish its imported marginal-tail lemma or the scanner's sublinear information bound.

4 Information Reset and Adaptive Patches

4.1 Why a one-sided filter estimate would be insufficient

In M535's own reset, let $\Psi = \Omega \otimes \Omega$ and let F swap a region R between copies; $\Phi = F\Psi$. A product purification from $I(R : C) \le b$ supplies a physical-buffer Hermitian contraction W. Writing $z = \operatorname{Tr}\rho_R^2$, the overlap lower bound gives $z \ge e^{-2b}$. The Gaussian intertwiner between $H^{(2)}$ and $FH^{(2)}F^*$ has

$$\|M\Psi - z\Phi\|\,, \quad \|M^*\Phi - z\Psi\| \le e^{-h\Delta^2/2}.$$

After truncation in time and Lieb–Robinson localization, both errors survive with a controlled additive loss. Dividing by z can cost e^{2b}; this cost is paid before choosing the collar width $C_p(1 + b + \log L)$. If D is the divided local operator, the two estimates control $(D^*D - I)\Psi$. The scalar inequality $(\sqrt{x} - 1)^2 \le (x - 1)^2$ for $x \ge 0$ then controls the polar replacement on this vector. A forward estimate alone cannot do this: the adjoint may amplify an error by $\|D\|$.

The local unitary leaves the two core reductions unchanged. Comparing to the swapped state yields a product core reduction, and entropy continuity costs $O(L^2)$ through the physical log dimension. The source chooses its inverse-power accuracy in advance to absorb this factor. No constant-dimensional entropy continuity estimate is used on an exponentially large physical space.

4.2 Conditional information is not monotone under conditioning

The conditional-cell estimate has cost $C(1 + \log s)^K$, with arbitrary conditioning unions of other cells. It cannot follow simply by replacing $I(G : C \mid B)$ with $I(G : C)$. The source orders tiles of

one family forward and the other backward; purity converts the second conditional quantity to the complementary system, where its earlier same-family tiles are separated. Holes around intersections and recursive scales $u' = 8u^\gamma$ give bounded branching and $O(\log\log s)$ generations. The final number of tiles is polylogarithmic. The angular argument resets the accumulated information after each dyadic layer. Its width budget is $C_3(1+\log s)^K + C_4 \log L$, dominated by geometric clearance for $s \geq D \log L$ once D and then the volume threshold are fixed. These are substantive geometric/analytic steps, not consequences of ordinary two-point clustering alone.

4.3 The noncommuting variational patch construction

For $u \leq B_0 \log L$ the source seeks a projector nearly fixing Ω, decomposed into cylinders at nested radii $u \leq r_j \leq 2u$. Let $m = \lfloor u/2 \rfloor + 1$, $a_j = \kappa/m$, $b = e^{-R}$, and minimize

$$N(R) = \min_{x_j \geq 0,\ \mathrm{Tr}\, x_j = 1} \|L_m \cdots L_1 \Omega\|\,, \qquad L_j = (x_j + bI)^{-a_j/2}, \qquad \phi = N(R)^{-1} L_m \cdots L_1 \Omega. \tag{4}$$

The feasible sets are finite-dimensional compact sets and $b > 0$, so minimizers exist. The filters are invertible; they are not mutually commuting. Stationarity under unitary variations, descending from the outermost radius, gives $[\rho_j, L_j] = 0$ for the relevant reduced state of ϕ. The diagonal constrained variations then give

$$x_i + b = \max(p_i/\lambda, b), \qquad p_i/(x_i+b) = \lambda\ (x_i > 0), \qquad p_i/b \leq \lambda\ (x_i = 0), \tag{5}$$

with $\lambda \leq 1$. This is a boundary KKT relation, not an assumption that a minimizing density is strictly positive.

The energy account strips outer filters using the proved commutators, not by commuting arbitrary nested operators. An edge crosses at most one selected radius. Pairing Hermitian matrix entries yields a coefficient bounded by $a_j^2(p_i + p_k)/2$; a q^2-term edge expansion and Cauchy–Schwarz give an energy increase at most $36q^2 J\kappa^2$. The cancellation is quadratic in a_j, so summing the $O(u)$ boundary edges and the m radii remains bounded. Choosing κ sufficiently small relative to Δ makes ϕ close to Ω.

Here the imported area law is used at constant accuracy, not inverse-power accuracy. An entropy cutoff gives a projector Q_j of rank $\exp(C(1+u))$ with a fixed small tail for Ω and hence for ϕ. With $T_j = \rho_j b (x_j + bI)^{-1}$, $\mathrm{Tr}\, T_j \leq e^{-R + C(1+u)} + 1/8 \leq 1/4$ beyond a fixed linear threshold. For $g(R) = \log N(R)$ the minimizing envelope is locally Lipschitz. Comparison with any fixed minimizer bounds its upper derivative by $\sum_j (a_j/2)\,\mathrm{Tr}\, T_j \leq \kappa/8$. Integration controls the norm growth. Inverting the filters and expanding at the first retained head gives an all-tail error bounded by

$$\exp\left\{\frac{\kappa}{2}[C_0(1+u) + \log 2 - 3R/4]\right\}. \tag{6}$$

Thus $R = R_0 \log L$ can give L^{-p} accuracy while each inside head has rank at most e^R. Preceding tails lie inside that head's radius and cannot increase its inside rank.

Orthogonalization at a later radius uses the image $(I - P_{W_j})S_j$, not the intersection $S_j \cap W_j^\perp$. The image preserves the span together with the earlier cylinders and cannot increase its dimension. The resulting source projector is

$$P = \sum_{j,\ell} |v_{j\ell}\rangle\langle v_{j\ell}|_{X_j} \otimes I_{X_{\mathrm{out}} \setminus X_j}, \qquad P^2 = P = P^*, \qquad \|(I-P)\Omega\| \leq L^{-p}, \tag{7}$$

with orthogonal cylinders and $\sum_j d_j \leq \mathrm{poly}(L)$. The inside vectors may still be extremely complicated. For example, $|0\rangle\langle 0| \otimes I_{\mathbb{C}^D}$ has a single inside vector but rank D. Replacing the cylinder count by the outer projector rank would be a mathematical error; the source's encoder explicitly avoids it.

5 Exact Encoded Frames and Their Actual Domain

The Ram frame reformulation is useful here only if its physical domain is specified. For (7), define the tagged encoder

$$K = \sum_{j,\ell} |j,\ell\rangle_{\mathrm{tag}} \otimes (|0\cdots 0\rangle\langle v_{j\ell}|)_{X_j} \otimes I_{X_{\mathrm{out}}\setminus X_j}. \tag{8}$$

Raw registers are retained even when blanked; the radius is part of the tag. Orthogonal tags remove cross terms in K^*K.

Proposition 5.1 (Exact range and disjoint-hole loss)**.** *For (8), $K^*K = P$. Thus K is an isometry on $\operatorname{ran} P$ and zero on $\ker P$, KK^* is the orthogonal projector onto $\operatorname{ran} K$, and K^* is its inverse on that encoded range. For finitely many disjoint outer holes let $K_{\mathcal{F}}$ be their combined encoder and $P_{\mathcal{F}} = \prod_a P_a$. Then*

$$K_{\mathcal{F}}^* K_{\mathcal{F}} = P_{\mathcal{F}}, \qquad 1 - \|K_{\mathcal{F}}\Omega\|^2 = \langle \Omega, (I - P_{\mathcal{F}})\Omega\rangle \le \sum_a \|(I - P_a)\Omega\|^2 =: \delta. \tag{9}$$

If $\delta < 1$, the normalized encoded state decodes to $P_{\mathcal{F}}\Omega/\|P_{\mathcal{F}}\Omega\|$ and has phase distance from Ω at most

$$\sqrt{2(1 - \sqrt{1-\delta})} \le \sqrt{2\delta}. \tag{10}$$

Proof. Multiplying (8) by its adjoint on the left gives exactly (7). Hence $\|Kv\|^2 = \langle v, Pv\rangle$ and $K = KP$. It follows that $(KK^*)^2 = KPK^* = KK^*$, and the range assertions follow. Disjoint outer holes give commuting projectors and encoders on distinct tensor factors. Thus their product has the displayed Gram operator. For commuting orthogonal projections the joint spectral basis has entries zero or one, and pointwise $1 - \prod_a p_a \le \sum_a (1 - p_a)$. Taking the Ω quadratic form proves (9). Put $r = \|P_{\mathcal{F}}\Omega\| = \|K_{\mathcal{F}}\Omega\|$. The decoded unit vector has real overlap r with Ω, so its squared distance is $2(1-r)$. Since $r \ge \sqrt{1-\delta}$, (10) follows. □

The source's linear loss estimate $1 - r \le \sum_a \varepsilon_a$ is sufficient for its asymptotic argument. Equation (9) improves the state-specific account to squared errors when the outer holes are disjoint. It does not apply to an arbitrary sequence of overlapping patch projections. For those, the contraction telescoping estimate still gives a sum of errors.

Proposition 5.2 (Finite-model frame, perturbation, and aliasing)**.** *Let $U : \mathbb{C}^d \to \mathcal{H}_L$ be an isometry specifying a model space, $T = K_{\mathcal{F}}U$, and $L_V = U^*(I - P_{\mathcal{F}})U$. Then*

$$G = T^*T = I - L_V, \qquad (1 - \|L_V\|)I \le G \le I. \tag{11}$$

If $\|L_V\| < 1$, $D = G^{-1}T^*$ satisfies $DT = I$ and $\|D\| \le (1 - \|L_V\|)^{-1/2}$. For $f = Ux + r$, $r \perp \operatorname{ran} U$, and data $c = K_{\mathcal{F}}f + e$,

$$\|UDc - f\| \le \|r\| + \frac{\|K_{\mathcal{F}}r\| + \|e\|}{\sqrt{1 - \|L_V\|}}. \tag{12}$$

If $\widetilde{T} = T + E$, $\|E\| \le \eta < \sqrt{\lambda_{\min}(G)}$, its least singular value is at least $\sqrt{\lambda_{\min}(G)} - \eta$.

Proof. Compress $K_{\mathcal{F}}^*K_{\mathcal{F}} = P_{\mathcal{F}}$ by U. The leakage is positive and at most I, giving (11). An SVD gives $DT = I$ and the norm bound. Subtracting f from UDc gives $UD(K_{\mathcal{F}}r + e) - r$, proving (12). Finally $\left\|\widetilde{T}x\right\| \ge \|Tx\| - \eta\|x\|$ gives the perturbation statement for every x. □

These are the actual Gram/leakage identities in the private proofs, specialized to the source's partial isometries. They add no artificial coercive term. The model condition is essential: $P = \operatorname{diag}(1,0)$ and $\Omega = (1,0)$ have zero state loss but no positive lower frame bound on $\mathbb{C}^2$. A finite numerical Gram bound also cannot establish uniform invertibility on a larger model, much less on every volume.

6 Sharper Whole-Group Tensor Truncation

This is the main restricted improvement. It preserves the source's *whole-group* rule: all open legs belonging to one original tensor vertex stay together. Splitting such a group among arbitrary owners invalidates the nuclear-norm argument.

Lemma 6.1 (Whole-vertex nuclear control)**.** *Consider a finite tensor graph with no self-loop at an original vertex. Each vertex is a vector of Hilbert norm at most one; internal edges are coordinate pairings. Contract all internal edges, leaving a tensor Z. Then $\|Z\| \le 1$. For every bipartition of the open legs that does not split a vertex's group, the sum of Schmidt singular values of Z is at most one.*

Proof. Merge two vertices by contracting *all* their common edges at once. After reshaping, this is a matrix product AB^{T}, whose Frobenius norm is at most $\|A\|_F \|B\|_F$. All edges internal to the merged component have then been consumed. Repeating within each connected component and taking tensor products between components proves the norm bound, without requiring the graph to be a tree. For a chosen whole-group bipartition, assign closed vertices arbitrarily to the two sides. Contract only edges within each side to obtain A, B, with Frobenius norms at most one. The remaining flattening is AB^{T}, so its nuclear norm is at most $\|A\|_F \|B\|_F \le 1$ by Schatten Hölder. Transpose rather than adjoint in a coordinate pairing does not change the singular-value bounds. □

An identity bypass is not a unit Hilbert-space tensor: its Frobenius norm on a D-dimensional wire is $\sqrt{D}$. Such wires must be factored as private identity operators before the lemma is applied. The small-rewrite proof in the source explicitly makes this distinction; omitting it would reintroduce the very private-dimension factor being eliminated.

Lemma 6.2 (Sharp nuclear-budget tail)**.** *If $s_1 \ge s_2 \ge \cdots \ge 0$, $\sum_i s_i \le \tau$, and $k \ge 1$ is an integer, then*

$$\sum_{i>k} s_i^2 \le \frac{\tau^2}{4k}. \tag{13}$$

The constant is sharp uniformly over the sequence length.

Proof. If the tail is empty there is nothing to prove. Otherwise let $a = s_{k+1}$. The first k entries sum to at least ka, and each tail entry is at most a. Hence

$$\sum_{i>k} s_i^2 \le a \sum_{i>k} s_i \le a(\tau - ka) = \frac{\tau^2}{4k} - k\left(a - \frac{\tau}{2k}\right)^2.$$

Equality holds for $2k$ entries all equal to $\tau/(2k)$. □

Theorem 6.3 (Anisotropic whole-group truncation)**.** *Let $Z \in \bigotimes_{j=1}^{g} \mathcal{H}_j$ have norm at most one. Suppose its flattening at group j has nuclear norm at most τ_j. For each j, let Q_j project onto its first $k_j \ge 1$ left singular directions, using all directions if the space is smaller. Put $Z_{\boldsymbol{k}} = (\bigotimes_j Q_j)Z$. Then*

$$\|Z - Z_{\boldsymbol{k}}\| \le \frac{1}{2}\left(\sum_{j=1}^{g} \frac{\tau_j^2}{k_j}\right)^{1/2}. \tag{14}$$

In the product of these singular bases it has at most $\prod_j k_j$ coefficients, and their absolute sum is at most $\sqrt{\prod_j k_j}$. For the graphs of Lemma 6.1, uniform $k_j = k$ gives $\|Z - Z_{\boldsymbol{k}}\| \le \sqrt{g}/(2\sqrt{k})$.

Proof. Each Q_j acts on a distinct tensor factor, so the projectors commute. The same projection union inequality used in (9) gives

$$\|Z - Z_{\boldsymbol{k}}\|^2 \le \sum_j \|(I - Q_j)Z\|^2 = \sum_j \sum_{i>k_j} s_{j,i}^2.$$

Apply Lemma 6.2 to each flattening. The product-basis coefficients $c_{i_1\dots i_g}$ have squared sum $\|Z_{\boldsymbol{k}}\|^2 \le 1$. Cauchy–Schwarz on at most $\prod_j k_j$ entries proves the absolute-sum bound. □

The source's stated $g/\sqrt{k}$ truncation follows by a triangle telescoping argument and a looser one-cut tail estimate. The improvement above uses orthogonality, not a new tensor factorization theorem. It also replaces the crude coefficient bound k^g by $k^{g/2}$. Since later Gaussian budgets depend on powers of the coefficient sum, this refinement has a concrete downstream use even when g is a fixed geometric constant. It does not prove a better universal PEPS exponent without an explicit accounting of all other source exponents.

6.1 Rank allocation and operator branches

For $\varepsilon > 0$, if all $\tau_j > 0$ and no dimension cap is active, minimizing $\prod_j k_j$ continuously subject to $\sum_j \tau_j^2/k_j \le 4\varepsilon^2$ equalizes τ_j^2/k_j. Indeed set $a_j = \tau_j^2/k_j$; maximizing $\prod_j a_j$ at fixed sum is AM–GM. Thus

$$k_j = \frac{g\tau_j^2}{4\varepsilon^2} \tag{15}$$

is the unconstrained allocation; taking $\lceil \max(1, k_j) \rceil$ preserves the error budget. If a group exhausts its dimension its tail is zero and it should be removed before reallocating. If any flattening has $\tau_j = 0$, then $Z = 0$ and no approximation is needed.

For a branch operator whose nontrivial tensor is Z and whose remaining wires are identity operators, the operator norm of the approximation error is at most the Hilbert norm in (14); tensoring a bypass identity does not change that operator norm. Suppose a rewrite has $M = \sum_{\nu=1}^{N} c_\nu M_\nu$ with such branches. Then

$$\left\|M - \widehat{M}\right\| \le \frac{1}{2} \sum_\nu |c_\nu| \left(\sum_j \frac{\tau_{\nu j}^2}{k_{\nu j}} \right)^{1/2}, \tag{16}$$

$$B_{\text{terms}}(\widehat{M}) \le \sum_\nu |c_\nu| \sqrt{\prod_j k_{\nu j}}.$$

Here B_{terms} is the absolute coefficient sum of the explicit normalized-resource/local-contraction expansion, not an assertion that the operator norm equals that sum. If $\|M\| \le 1$ and the first bound is η, then $\widehat{M}/(1+\eta)$ is a contraction and is within 2η of M, since $\left\|\widehat{M}\right\| \le 1 + \eta$.

7 Rewrites, Pair Effects, and Density Compression

7.1 The exact rewrite before approximation

For old and new encoded layouts the source first defines the actual map

$$M = K_{\text{new}} P_m \cdots P_1 K_{\text{old}}^*. \tag{17}$$

It is a contraction without any truncation. Each patch inner square must cover the required changed raw sites and entire affected outer holes. Each patch outer square, after removing old hole interiors,

meets at most two old owners; the analogous condition holds separately for new owners. The two owner lists need not coincide. Merely checking hole centers is insufficient for this support condition.

On the reference state, telescoping gives an error at most the sum of the old-hole and patch errors. The telescoping identity needs no commutativity: in a product $A_s \cdots A_1 - I$, insert $A_s \cdots A_2$, then $A_s \cdots A_3$, and continue. Each difference is a later contraction product times $(A_j - I)$ on the original vector.

After expanding tags and patch cylinders, an old decoder ket is fully intercepted by a later patch bra; a new encoder bra is intercepted by an earlier patch ket. Blank vectors, tags, and private bypass identities are factored out. The surviving nonprivate tensors have boundedly many normalized vertices. Patch bras form whole input groups on at most two old owners; patch kets form whole output groups on at most two new owners. Lemma 6.1 now applies. A tensor spanning two parties is a normalized pair resource or its effect; a one-party tensor is a local contraction. Equation (16) improves this exact step without replacing (17) by a surrogate.

7.2 Common garbage for effects

Let η be a unit bipartite vector. The source appends $m-1$ copies of η and averages cyclic permutations of the m joint registers. Call the resulting isometry-average T_m. It sends η to $\eta^{\otimes m}$. If $v \perp \eta$, the m vectors with v in one position and η elsewhere are mutually orthogonal, so

$$\|T_m - |\eta^{\otimes m}\rangle\langle\eta|\| \le m^{-1/2}. \tag{18}$$

The cyclic permutation is the product of a local permutation at each party. For a gate $\sum_\xi c_\xi M_\xi$ containing at most r effects per monomial, the error is at most $rm^{-1/2}\sum_\xi |c_\xi|$, with at most m^r times as many terms.

Every occurrence in every branch receives its own resource stack. An inactive branch appends the same complete stack that an active branch would leave as ideal garbage. Therefore the ideal output garbage is independent of the branch, and coherent cross terms are preserved. Replacing an effect separately without this common-garbage construction would not justify tracing out its register after summing branches. Once a gate is approximated to error η it is divided by $1+\eta$; subsequent circuit estimates consequently still use contractions.

7.3 The dimension-free Gaussian calculation

Write two unit bipartite sources in Schmidt form with probability vectors λ, μ. For their rank-one cross operator $X = |\eta\rangle\langle\widetilde{\eta}|$, the source samples complex standard Gaussians g_{ac} and forms

$$U = \sum_{a,c} (\lambda_a \mu_c)^{1/4} g_{ac} |u_a\rangle\langle\widetilde{u}_c|, \tag{19}$$

$$V = \sum_{b,d} (\lambda_b \mu_d)^{1/4} \overline{g_{bd}} |v_b\rangle\langle\widetilde{v}_d|, \qquad \widehat{X} = \frac{1}{k}\sum_{j=1}^{k} U_j \otimes V_j.$$

Then $\mathbb{E}\widehat{X} = X$. For $z_{\alpha\beta} = g_\alpha \overline{g_\beta} - \delta_{\alpha\beta}$, complex Wick pairing gives

$$\mathbb{E} z_{\alpha\beta}\overline{z_{\gamma\delta}} = \delta_{\alpha\gamma}\delta_{\beta\delta}. \tag{20}$$

Thus a centered sample average has diagonal covariance $1/k$. No bound on the norm of an individual Gaussian realization is assumed.

For a corrected slot set S, let A be its endpoint parties. Then $|A| \le 2|S|$. Only gates touching A are expanded into their monomials; the exterior gates remain whole contractions. After fixing endpoint

physical ket and bra indices, the endpoint maps $A_x, \widetilde{A}_y$ have a common garbage output, so $O = \widetilde{A}_y^* A_x$ is a contraction. Across A and its exterior, all source slots are uncorrected, because every corrected slot has both endpoints in A. They provide normalized density weights τ and $\widetilde{\tau}$ on the two sides.

For completeness, the essential weighted estimate is as follows. If $R, \widetilde{R}$ are positive matrices of trace one and $\|O\| \le 1$, then

$$\left\| \widetilde{R}^{1/4} O R^{1/4} \right\|_2^2 \le 1. \tag{21}$$

Indeed Schatten Hölder with exponents $4, \infty, 4$ gives a bound by $(\operatorname{Tr} \widetilde{R})^{1/2} \|O\|^2 (\operatorname{Tr} R)^{1/2} = 1$ after squaring. Here $R = \operatorname{diag}(p_i) \otimes \tau$ and $\widetilde{R} = \operatorname{diag}(\widetilde{p}_l) \otimes \widetilde{\tau}$, where $p, \widetilde{p}$ are products of the Schmidt probabilities of corrected slots. They remain probability distributions, whatever their dimensions. Equation (20), followed by (21), bounds the relevant second moment by $\prod_{e \in S} k_e^{-1}$.

To convert it to trace norm, the source first sandwiches the interior matrix by the square roots of $\tau, \widetilde{\tau}$ and applies Schatten Hölder with exponents $4, 2, 4$. Exterior contractions and the partial trace cannot increase this trace norm: duality gives $|\operatorname{Tr}[(Y \otimes I) K Z \widetilde{K}^*]| \le \|Y\| \, \|Z\|_1$ for contractions $K, \widetilde{K}$. This argument works for non-Hermitian Z as well. The resulting first-moment error per fixed monomial/output choice is at most $\prod_{e \in S} k_e^{-1/2}$. This is the place where a naive dimension factor would have ruined the theorem; the normalized fourth-root weights and common garbage remove it.

7.4 A precise nonuniform resource extension

The following result separates a proved allocation rule from the source hypotheses needed to invoke the dimension-free subset estimate.

Theorem 7.1 (Slot-dependent Gaussian budget)**.** *Suppose a finite contraction network satisfies the preceding source-only description, the common-garbage condition, and bounded nonprivate participation. Let $B_h \ge 1$ bound the absolute coefficient sum of gate h, and let $d_P \ge 1$ bound the physical output dimension of party P. For a slot e with endpoint set A_e, define*

$$Q_e = \prod_{h:\ h \text{ touches } A_e} B_h^2 \prod_{P \in A_e} d_P^2. \tag{22}$$

Use independent samples at distinct slots, with $k_e \ge 1$. Under the dimension-free subset calculation just described, the output operators satisfy

$$\mathbb{E} \, \|\widehat{\rho} - \rho\|_1 \le \prod_e \left(1 + \frac{Q_e}{\sqrt{k_e}}\right) - 1 \le \exp\left(\sum_e \frac{Q_e}{\sqrt{k_e}}\right) - 1. \tag{23}$$

The operators need not be positive. With no slots the error is zero and no allocation is needed. Otherwise fix $\varepsilon > 0$, set $t = \log(1 + \varepsilon) > 0$ and $W = \sum_e Q_e^{2/3}$, and take the integer choice

$$k_e = \left\lceil (W/t)^2 Q_e^{2/3} \right\rceil \tag{24}$$

which makes the expectation at most ε. Before rounding, the positive real allocation minimizes $\sum_e k_e$ subject to the additive sufficient budget $\sum_e Q_e / \sqrt{k_e} \le t$, without lower-bound constraints. No optimality for the product bound or the actual expected error is asserted.

Proof. Expand the multilinear difference by its nonempty corrected slot set S. Its endpoint set is $A = \bigcup_{e \in S} A_e$. Expanding ket and bra monomials at each gate touching A costs $\prod_{h \text{ touches } A} B_h^2$; physical ket/bra choices cost $\prod_{P \in A} d_P^2$. Since every factor is at least one, these costs are at most $\prod_{e \in S} Q_e$,

even if a gate or party is counted more than once. The preceding covariance/witness calculation and the triangle inequality therefore give

$$\mathbb{E}\,\|F_S\|_1 \le \prod_{e\in S} \frac{Q_e}{\sqrt{k_e}}.$$

Summing over all nonempty subsets proves the product formula; $1+x \le e^x$ proves the second inequality. The proposed real allocation has $\sum_e Q_e/\sqrt{k_e} = t$, and rounding upward preserves the inequality. To prove optimality, Hölder gives

$$W = \sum_e (Q_e/\sqrt{k_e})^{2/3} k_e^{1/3} \le \left(\sum_e Q_e/\sqrt{k_e}\right)^{2/3} \left(\sum_e k_e\right)^{1/3}.$$

Thus $\sum_e k_e \ge W^3/t^2$, with equality for the stated real values. If the lower constraint $k_e \ge 1$ is active, the ceiling still gives a valid allocation, but is not claimed to solve the constrained integer optimization exactly. □

If each party touches at most b gates, with $B_h \le B$ and $d_P \le d$, then $Q_e \le B^{4b} d^4$, recovering the source's uniform bound. For $N \ge 1$ slots and $0 < \varepsilon \le 1$, one may take $k \ge (8NQ/\varepsilon)^2$ and obtain its inverse-power error allocation. Equation (16) lowers the admissible B_h for whole-group rewrites. This is a rigorously specified improvement to a resource bound, not a measured runtime gain. Markov's inequality can give existence of a suitable realization; it does not identify it efficiently or bound its bit precision.

8 Ownership Geometry and Global Assembly

The geometry is an essential hypothesis of the compression, not decorative bookkeeping. Dyadic blocks on a padded grid N with $L \le N < 2L$ receive parties at each scale. A block party is active only for its own scale, its child-scale repaint, and the bounded transition operations. Fresh auxiliary sheets are retired after their use; their parties are not silently reused for every later scale. This is the source's mechanism for bounded lifetime participation.

At scale n put $t = D \log L$ and $h_n = \varepsilon_0 \min(n, t)$. Holes occur at true junctions of at least three actual owners; the outer squares are disjoint. Placeholder owners outside the physical grid hold no persistent raw registers or tags. A tag belongs to an incident actual owner. Identifying two owner names may remove an obstruction, but must not create a new third-owner condition.

8.1 Bulk bands and point repairs

For large blocks an edge uses the normalized transverse coordinate $x = d/[10^{-3} \min(s, n-s)]$. Its taper is important near endpoints. The auxiliary strip word is

$$C \mid B \mid A \mid C \mid A \mid B \mid C$$

with boundaries at $-6, -5, -4, -3, -2, -1$. The exchange strip has boundaries -7 and -3.5, both lying in the common C region. After exchange the main sheet has word $C \mid B \mid A \mid C \mid A \mid B$ with boundaries $-6, -5, -4, 0, 1$. The subsequent deaths and birth remove the inner C, then A, restore the intended C band, and remove the outer B band. These are ownership changes of reference-state representations, not instantaneous physical swaps between remote owners.

Births are implemented using a product-purification isometry on the buffer owned by one party. A reverse birth contains a pair effect. For an exchange, sites on two copies are renamed privately, a

private compensating swap is made on common-owner registers, and a purification swap is inserted in the correct order. A whole hole must lie on one side of the exchange boundary. Its tags are renamed with its raw sheet. The source's error estimate is of order $\sqrt{I(T : E)}$; it is not an operator-norm estimate on every input state.

Near finitely many endpoints and crossings, logarithmic point squares are homogenized to a surrounding owner before a bulk operation and restored afterward. The small-patch cover must cover entire old and new hole footprints. The source uses compact families of scaled patterns with positive hole radii and no triple meeting outside hole interiors. This is more than drawing a finite list of centers. The parameter order is also important: band templates, a large scale cutoff, hole fraction, patch mesh, clearance, information constant D, and finally patch-rank constants are selected in that order. Retuning an earlier geometric constant using a later rank exponent would create a circular argument.

The full geometry file was read, including arbitrary boundary clipping, small scales, tag ownership, and sheet retirement. This report does not claim a machine-certified exhaustive proof of all continuous pattern families. Its local algebra assumes the stated support/owner conditions, which should be checked on any separately implemented pattern library.

8.2 Errors and routing

The source uses at most CL^3 changes, each with error CL^{-20}, to reach a vector near $\Omega \otimes \Xi$; the auxiliary reference norm is retained, not normalized away as an assumed success probability. It obtains a physical density error CL^{-10} before distributed compression. The latter is applied with inverse-power accuracy such as L^{-5}. Constants are absorbed only after a sufficiently large volume threshold; a finite collection of small volumes is handled by an exact representation with a larger constant prefactor.

At dyadic scale $n = 2^j$, nonunit party anchors are at $(nr + n/2, ns + n/2)$. The row coordinate has two-adic valuation $j - 1$. Routing first horizontally and then vertically therefore assigns different nonunit scales to different rows/columns. Only equal or adjacent scales are linked and each link has length $O(n)$; bounded degree and locality then give constant edge congestion. Without this valuation separation, stacking $O(\log L)$ scales on the same edge could give quasipolynomial rather than polynomial bonds. Coordinate folding into the original grid uses $f(x) = x$ for $x \le L - 1$ and $f(x) = 2(L - 1) - x$ otherwise. For the padded range and $L \ge 2$ it stays in the physical interval; the preimage multiplicity of a physical edge is bounded. Independent virtual indices are carried through crossings. This routing conclusion is conditional on the bounded-degree, adjacent-scale, short-link graph provided by the owner construction, not on an arbitrary network graph.

9 Column Extraction Without Positivity

The Gaussian network is an operator σ, not necessarily positive or Hermitian. Requiring positivity after the fact would add an unproved operation. Fortunately the source's column method does not need it.

Proposition 9.1 (A sharp column normalization bound)**.** *Let $\|\Omega\| = 1$ in a finite tensor-product space, and let $\|\sigma - P_\Omega\|_1 \le \eta < 1$. In any fixed product basis there is a column $|z\rangle$ with $a_z = \langle\Omega|z\rangle \ne 0$ such that $w = \sigma|z\rangle/a_z$ satisfies $\|w - \Omega\| \le \eta$. It is nonzero and*

$$\min_{|\zeta|=1} \|\zeta w/\|w\| - \Omega\| \le \sqrt{2(1 - \sqrt{1 - \eta^2})}. \tag{25}$$

The bound is sharp as a statement about a vector in the radius-η ball about Ω. Fixing the input column in an operator network does not increase its virtual bond dimensions.

Proof. Put $E = \sigma - P_\Omega$. Since Schatten norms satisfy $\|E\|_2 \le \|E\|_1$,

$$\sum_z \|E|z\rangle\|^2 \le \eta^2, \qquad \sum_z |a_z|^2 = 1.$$

If every nonzero a_z had $\|E|z\rangle\| > \eta|a_z|$, summing would contradict the first inequality. Thus the desired column exists and $w - \Omega = E|z\rangle/a_z$. Its norm is positive since $\eta < 1$. The distance from Ω to the complex line spanned by $v = w/\|w\|$ equals $\sqrt{1 - |\langle\Omega, v\rangle|^2}$ and is at most $\|w - \Omega\| \le \eta$. Hence $|\langle\Omega, v\rangle| \ge \sqrt{1-\eta^2}$, proving (25). For sharpness, choose a unit v with real overlap $\sqrt{1-\eta^2}$ and let $w = \sqrt{1-\eta^2}v$. Then $\|w - \Omega\| = \eta$ and equality holds. Finally inserting the fixed local basis indices in the input legs changes local tensors, not the network's virtual index sets. □

This improves the source's sufficient bound 2η without changing its qualitative conclusion. There is no lower bound on $|a_z|$. Consequently the division is existential and may be numerically unstable. For a useful but nonalgorithmic variant, sample z on the support $|a_z| > 0$ with probabilities $|a_z|^2$. For $s > 0$,

$$\Pr\{\|E|z\rangle\|/|a_z| > s\} \le \eta^2/s^2. \tag{26}$$

This follows by Markov from the same squared sum. Sampling this distribution presupposes access to the unknown ground state; it does not resolve the construction problem.

10 Physical Metrics and Honest Candidate Checks

If candidate vectors are represented in a nonorthogonal finite synthesis $J : \mathbb{C}^d \to \mathcal{H}_L$, injective on its chosen model, the physical metric is $G = J^*J > 0$, not the Euclidean coefficient norm. Put

$$P_J = JG^{-1}J^*, \qquad K_H = J^*HJ, \qquad B = G^{-1}K_H.$$

The private physical-metric proof specializes exactly to this setting.

Proposition 10.1 (Residual and ground-state branch)**.** *For every c and scalar λ,*

$$\|(H-\lambda)Jc\|^2 = \|(B-\lambda)c\|_G^2 + \|(I-P_J)HJc\|^2. \tag{27}$$

For a normalized physical vector ψ, under (1),

$$\min_{|z|=1} \|z\psi - \Omega\| \le \sqrt{2\langle\psi, (H-E_0)\psi\rangle/\Delta}. \tag{28}$$

If additionally $\lambda \in [E_0, E_0 + \Delta/2]$ and $r = \|(H-\lambda)\psi\|$, the phase distance is at most $2\sqrt{2}r/\Delta$.

Proof. The two orthogonal components of $(H-\lambda)Jc$ are $J(B-\lambda)c$ and $(I-P_J)HJc$, proving (27). Write $\psi = a\Omega + \xi$, $\xi \perp \Omega$. The gap gives $\Delta\|\xi\|^2 \le \langle\psi, (H-E_0)\psi\rangle$. The squared phase distance is $2(1-|a|) \le 2(1-|a|^2) = 2\|\xi\|^2$, giving (28). In the stated spectral interval every excited energy is at least $\Delta/2$ from λ, so $r^2 \ge (\Delta^2/4)\|\xi\|^2$, proving the last bound. □

A zero projected residual can coexist with a nonzero omitted component. Even a zero *full* residual does not select the ground state without a branch condition: an exact excited eigenvector has zero residual at its own eigenvalue. Neither the nonpositive compressed operator nor an unverified PEPS contraction supplies the positive energy evaluation required in (28). Computed bounds must include physical normalization, omitted-space error, and rigorous evaluation error if they are to be certificates rather than numerical diagnostics.

11 Gap Audit, Benefits, Costs, and Further Work

11.1 Invalid shortcuts excluded by the analysis

An entropy bound on a logarithmic square does not directly give the needed inverse-power rank tail. For example, take a spectrum $(1-\theta, \theta/D, \dots, \theta/D)$ with $D = \lceil \exp((\log L)^2) \rceil$ and $\theta = 1/\log L$ for large L. Its entropy is $h(\theta) + \theta \log D = O(\log L)$, yet the mass after any fixed polynomial number of entries is asymptotic to $1/\log L$. This is a spectral example, not a gapped-Hamiltonian counterexample. It explains why the adaptive noncommuting filter is doing necessary work.

State-specific encoder accuracy is not a global frame inequality; inside-cylinder rank is not outer-space rank; groupwise nuclear control does not survive arbitrary splitting of groups; identity wires do not have unit Hilbert norm; conditional mutual information is not monotone under adding conditioning; bounded participation per scale is not bounded participation over a lifetime; and a nonpositive operator approximation is not itself a quantum state. The report has retained these distinctions at the exact points where the source uses them.

No explicit counterexample to the pinned M535 theorem has been established here. The full companion area-law/collar chain is still an unverified global dependency. The geometry's uniform continuous families and the interface between all analytic constants also lack an independent formal verification in this report. The complete source read supports a detailed dependency account, not a whole-proof certification.

11.2 Benefits of the restricted reformulation

The exact encoder Gram operator makes the recoverable subspace and its lost directions visible. For disjoint holes, a squared-error union bound is sharper than a linear norm loss. The sharp singular-tail estimate and commuting group projections reduce the whole-group truncation cost, while the coefficient absolute-sum bound propagates into the actual Gaussian budget. Nonuniform slot allocation uses known heterogeneous costs instead of automatically applying the worst one everywhere. The column lemma keeps the density compression useful without inventing positivity. Physical metrics prevent coordinate conditioning from masquerading as small physical error. These benefits follow from the proved formulas, not from benchmarks or new asymptotic complexity claims.

11.3 Costs and limitations

Computing the optimal group singular spaces can itself involve very large private dimensions. The source's exponents depend on gap, filter order, geometric clearances, and repeated inverse-power requirements. The nearly unit collar exponents hide potentially enormous thresholds. The Gram lower bound can deteriorate on an enlarged model; canonical decoding may destroy locality. Gaussian compression loses positivity and gives no stable column denominator or precision budget. Preparing exact private ground-state copies makes the construction unsuitable as a discovery algorithm without further ideas. None of the local improvements removes these costs or the companion dependencies.

11.4 Concrete extensions and their obstacles

1. Complete an independent proof audit of the companion marginal-tail, scanner, transport, and replica estimates. The obstacle is preserving uniform constants through noncommuting spectral cutoffs and irregular collars, not merely checking the positive replacement in isolation.

2. Extract an explicit finite library of owner patterns with interval enclosures for minimum clearances and complete support coverage. Finite samples of a continuous pattern family are insufficient; a compactness argument needs a quantified positive separation and all degeneracies.

3. Combine (16) and (24) with the actual branch counts to report explicit exponents. This requires all gate coefficient costs and the fixed lifetime bound, not just the improved local truncation constant. A smaller certified budget is not automatically a faster implementation.

4. Replace exact private copies and existential column selection by a constructive procedure. This would need an independently accessible state preparation or ground-space approximation and a test for good samples; estimating the relevant global norms may itself be difficult.

5. Extend to degenerate ground spaces only with a new information and branch-selection argument. The present purity dualities and rank-one column lemma use a unique pure state. A mixed ground-space projector is not a drop-in replacement, even though [5] provides relevant conditional mixed-state methods.

6. Extend to other lattices or long-range interactions by rebuilding the collar geometry, locality tails, and constant-congestion routing. Replacing a square by a different graph without controlling those three features does not preserve the polynomial claim. Allowing a gap that shrinks with L also changes constants into size-dependent quantities and may invalidate the polynomial exponent.

12 Contribution and Evidence Contract

The realized contribution is C2: complete, limited refinements of exact source operators, with conservative resource and domain accounting. The independent proofs are Propositions 5.1, 5.2, 9.1, 10.1, Lemmas 6.1 and 6.2, and Theorems 6.3 and 7.1. The slot theorem explicitly imports the source's network witness hypotheses. The underlying ingredients are elementary projection theory, Schatten inequalities, singular-value truncation, and constrained error allocation; no first-in-literature claim is attached to them.

The sidecar check script tests exact rational and finite floating-point instances of these statements, including counterexamples to invalid shortcuts. It records tolerances, seeds, sizes, outcomes, and the test script hash. Such checks can reveal implementation or algebra errors; they cannot certify the companion theorem, all owner geometries, all volumes, or an asymptotic PEPS exponent. The written local proofs, the conditional global account, and finite tests are distinct evidence classes.

Provenance records preserve the pre-read hashes for the M535 build texts, actual read ranges, current-byte comparisons, and a nested-path SHA-256 freeze. The batch source-identity record confirms that all fifteen source text files, including the text outside the build tree, match both captured October 8 Git trees. That dated cache comparison is not a new live-branch fetch or a proof check, and does not establish continuous byte stability throughout reading. The physical export and every final page's visual inspection are recorded in the release sidecar. Native-editor dispatch and any platform limitation are recorded separately; physical export is not described as native success.

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